Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/12/2021
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Capillary Flows and Draining in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	607-255-4749
Organization Name:	Cornell University		
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PI Address 2:	120 Olin Hall		
PI Web Page:			
City:	Ithaca	**State:**	NY
Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	01/31/2019	**End Date:**	04/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
Contact Email:	tyler.r.hatch@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 4/30/2022 (previously extended to 12/31/2021) with Olivier Desjardins as PI, overseeing Josh McCraney's research after Paul Steen passed away (Ed., 12/2/21)		
Key Personnel Changes/Previous PI:	NOTE (Ed., November 2020): Principal Investigator Dr. Paul Steen passed away in September 2020; Dr. Olivier Desjardins will now oversee Joshua McCraney's research on this project. NOTE (Ed., Mar 2019): This work is result of proposal by graduate student Joshua McCraney, Ph.D. candidate at Cornell University, who will be performing the work with advisor PI Dr. Paul Steen.		
COI Name (Institution):	McCraney, Joshua Ph.D. (Graduate Student (PostDoc May 2021) PI: Cornell University)		
Grant/Contract No.:	80NSSC19K0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Capillary Flow Experiments (CFE) for Interior Corner Flows (ICF) were conducted aboard the International Space Station (ISS) and rely solely on video for high-quality quantitative data. We propose to digitally measure the still video images of the time-dependent interface height. Since container geometry is known, the meniscus heights can be used to determine volumetric drain rates. This data would then serve as both a benchmark for current theory on drain rates and a guide for extending capillary flow models to a diverse set of geometries and flow conditions. We also propose to utilize and extend a numerical tool to simulate 3D transient corner flows with and without topological change. To pursue any discrepancy between data and numerics, supportive ground-based experiments at micro-scales and low-gravity experiments conducted in the Dryden Drop Tower at Portland State University are proposed. Understanding the dynamics of capillary flows and responses of drain flows to various geometries offers design advantages for spacecraft waste-management systems, fuel control, hydration containment, and cryogenic flows. Capillary flows and drain problems are relevant to a broad range of two-phase flow technologies in low gravity, including 3D vane networks for positioning of fuels, an important problem concerning propellant liquid management aboard spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The capillary drain flows see immediate terrestrial application for biofluids during lab-on-chip processing. For low-gravity environments, capillary-dominated fluid behavior is common over large length scales in the near-weightless environment aboard spacecraft. In part due to both the expense and limited opportunities for designers to develop significant intuition of general multiphase phenomena in such environments, developing robust fluid systems for spacecraft fuels management, thermal control, and water processing for life support remains challenging. For the specific example of liquid fuels aboard orbiting spacecraft, capillary draining can serve as a limit to the life of the spacecraft if and when the residual fuel in the tank is or becomes inaccessible. It is essential to establish the maximum capillary flow rate at which a container in a micro-gravity environment can be drained to increase satellite predictability and lifespan (when fuel is empty, the satellite is essentially expired).
Task Progress:	There have been three significant accomplishments since the last reporting period: two publications in peer reviewed journals and successful defense of the PhD dissertation. The first publication is theoretical, and accomplishes two feats: 1) corrects a misunderstanding widespread in the spectral theoretical field and 2) proposes an efficient and accurate algorithm that identifies the resonant frequencies for a myriad of fluid systems. The second publication is experimental, and accomplishes two feats: 1) reduces liquid drain video data from the ISS Physical Sciences Informatics (PSI) archive into easy-to-use data files and 2) compares reduced data to a simple theoretical prediction, identifying regimes where the theory is valid and the limitations thereof. The PhD dissertation consolidates these two papers, as well as a third publication on numerical modeling that was reported in the previous annual report. Since receiving the PhD, McCraney has continued this work as a postdoctoral researcher, continuing at Cornell University.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Articles in Peer-reviewed Journals	McCraney J, Weislogel M, Steen P. "Resonant mode scanning to compute the spectrum of capillary surfaces with dynamic wetting effects." J Eng Math. 2021 Jul;129:10. https://doi.org/10.1007/s10665-021-10150-2 , Jul-2021
Articles in Peer-reviewed Journals	McCraney J, Weislogel M, Steen P. "The draining of capillary liquids from containers with interior corners aboard the ISS." npj Microgravity. 2021 Nov 11;7(1):45. https://doi.org/10.1038/s41526-021-00173-5 ; PMID: 34764319 , Nov-2021
Dissertations and Theses	McCraney J. "Capillarity: Draining and Surface Vibrations." Dissertation, Cornell University, May 2021. , May-2021